Timeline of discoveries

Since ancient times, debate and experiment have led to discoveries that further human understanding of how the world works—but there are still many questions left to answer.

1890–PRESENT

BOMBE CODE-BREAKING MACHINE

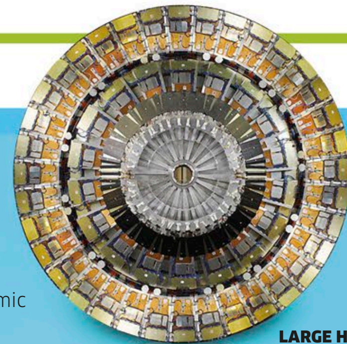
1936

Computer science
British code-breaker Alan Turing develops the first programmable computer, laying the foundations of modern computer science.

NUCLEAR EXPLOSION

1942

Nuclear energy
Italian–American physicist Enrico Fermi leads a US team that builds the world's first nuclear fission reactor. In 1945, the first atomic bomb is dropped on Hiroshima.



LARGE HADRON COLLIDER

2012

Higgs boson
The Higgs boson particle is identified, confirming the Standard Model of particle physics developed in the 1970s.

1700–1890

1847

Energy conservation
German physicist Hermann von Helmhotz states that energy cannot be created or destroyed, it can only change its form.

PERPETUAL MOTION MACHINE

1831

Electromagnetic induction
After electricity and magnetism are linked, English scientist Michael Faraday uses electromagnetic induction to generate electricity.

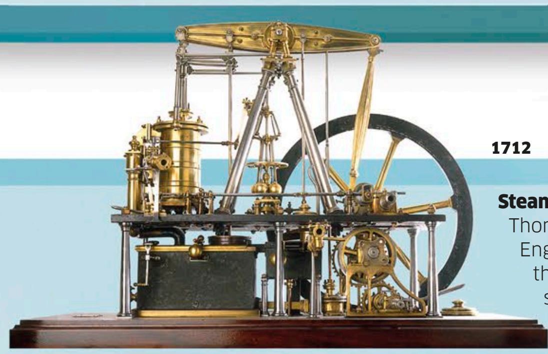
FARADAY'S COIL

1799

Current electricity
Italian inventor Alessandro Volta creates an electric current by stacking disks of zinc, copper, and cardboard soaked in salt water in alternate layers—the first battery.

VOLTAIC PILE

The Industrial Revolution
Scientific principles understood by the 18th century were applied to large-scale practical machines during the Industrial Revolution. The power of electricity was unlocked, which led the way for a surge in new technology.



NEWCOMEN ENGINE

1712

Steam engine
Thomas Newcomen, an English engineer, builds the world's first practical steam engine. It is followed by James Watt's more efficient engine and Richard Trevithick's steam locomotive.

LEYDEN JAR

1712

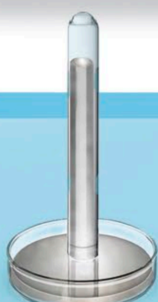
Static electricity
German scientist Ewald Georg von Kleist invents the Leyden jar, a device that can store a static electric charge and release it later.

1700–1890

1500–1700

1643

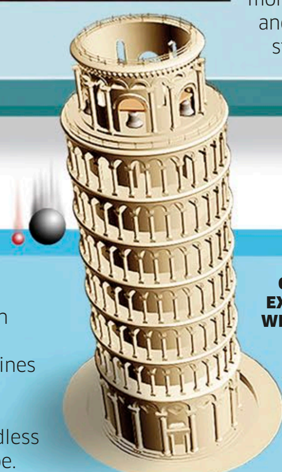
Atmospheric pressure
Italian physicist Evangelista Torricelli creates a simple barometer that demonstrates atmospheric pressure.



BAROMETER

1604

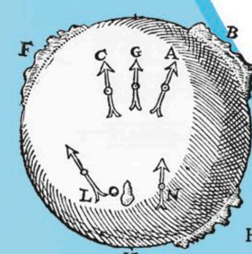
Falling bodies
In a letter to theologian Paolo Sarpi, Italian scientist Galileo outlines his theory that all objects fall at the same rate, regardless of mass or shape.



GALILEO'S EXPERIMENT WITH FALLING BODIES

1600

Earth's magnetism
English scientist William Gilbert theorizes that the Earth must have a huge magnet inside.



GILBERT'S MAGNET

16TH CENTURY

1300

Bending light
German monk Theodoric of Freiburg uses bottles of water and water droplets in rainbows to understand refraction.

A new age of science
The scientific revolution, from the mid-16th to the late 18th centuries, transformed understanding of astronomy and physics. This period saw the development of the scientific method of experiment and observation.

NICOLAUS COPERNICUS

1543

Solar system
Polish astronomer Nicolaus Copernicus states that the Earth and planets orbit around the sun.